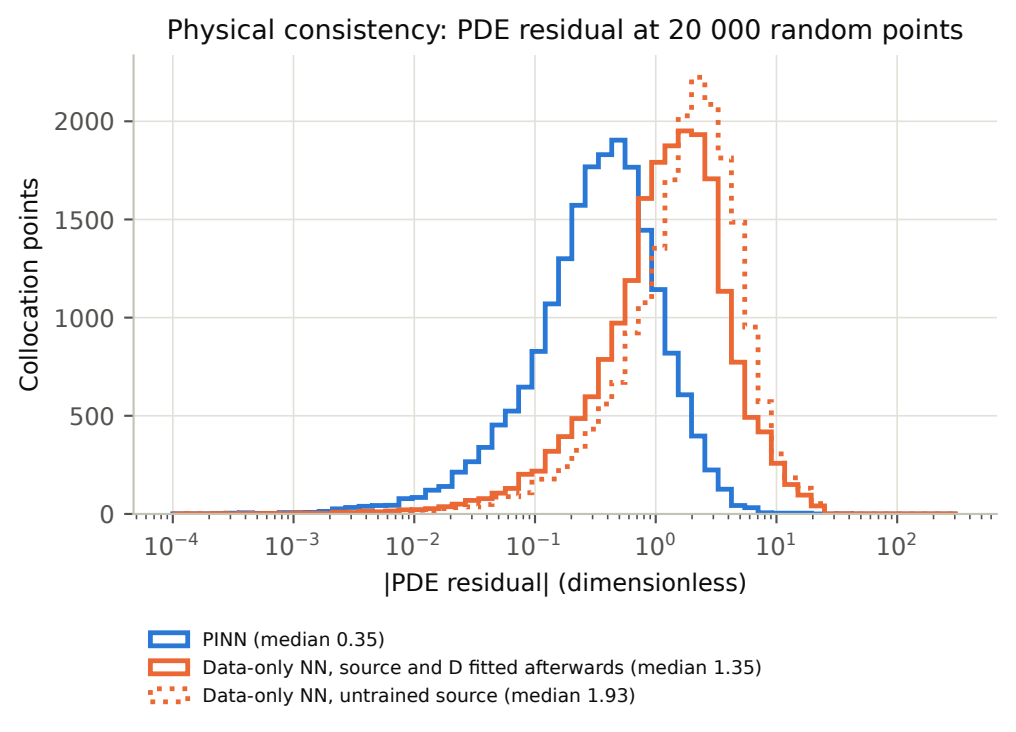
Physical consistency: PDE residual at 20 000 random points
Collocation points
2000
1500
1000
500
0
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
|PDE residual| (dimensionless)
PINN (median 0.35)
Data-only NN, source and D fitted afterwards (median 1.35)
Data-only NN, untrained source (median 1.93)